CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 4: Containerization with Docker & Kubernetes Basics

Course: Open Elective III [23OECS31]

Category: Assignments | Assignment #4 | Academic Year: 2026-27

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1. Overview & Objective:

Operating system-level virtualization, Docker containers, and container orchestration.

2. Assignment Problems & Instructions:

1. Explain Linux namespaces PID, NET, MNT and cgroups forming the foundation of containers.
2. Write an optimized multi-stage Dockerfile minimizing final image size for a Node.js/Java app.
3. Describe Kubernetes control plane components API Server, etcd, Scheduler, Controller Manager .
4. Define a Kubernetes Deployment and ClusterIP Service YAML manifest managing 3 replica pods.

Submission Guidelines & Verification:

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